LiftStats: Statistical Modeling, Biomechanical Clustering, and Longitudinal Progression for Strength Athletes

Technical Report: Data Science

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1. Project Overview & Motivation

Ask ten recreational lifters in any gym what counts as a “respectable” bench press, and almost all will quote arbitrary milestones like 100 kg or “two plates” (225 lb). However, real human physical strength does not scale in round numbers. Force production is inherently non-linear, governed by an intricate combination of biological sex, chronological age, body mass, and skeletal leverage. Traditional online strength tables rarely scale properly across bodyweights, while fitness social media popularizes standards skewed by elite genetics or undisclosed performance-enhancing drugs. For ordinary gym-goers, this breeds distorted expectations, chronic overtraining, and severe joint injuries from chasing arbitrary targets.

To replace this guesswork with empirical data, our team built LiftStats, an open-source statistical benchmarking platform and progression-forecasting engine trained on 1,560,406 validated competition records from OpenPowerlifting. We organized our implementation into three vertical slices: Pavlo Vus engineered the data ingestion pipeline, data cleaning rules, and the non-parametric percentile engine; Gergo Lindemann developed the unsupervised biomechanical archetype clustering and the drug-tested federation comparison toggle; and Zhishang Chen built the longitudinal progression regression model, the bodyweight normalization engine (Wilks and DOTS), and the shareable frontend cards. All components were locked to a unified schema contract codified in our project specification (FULL_CONTEXT.md).

A foundational design objective for LiftStats was injury prevention. Naive online progression calculators assume linear gains indefinitely - promising 2.5–5 kg increases every month. In biological reality, human muscular adaptation exhibits steep diminishing marginal returns. When progress naturally slows, linear advice pushes athletes into reckless “ego lifting,” triggering acute tendon tears and lumbar injuries. By modeling diminishing returns mathematically and surfacing an honest 80% empirical prediction interval [q10, q90], LiftStats normalizes natural plateaus and guides lifters toward safe, sustainable periodization. Beyond powerlifting, our methods - continuous simplex modeling of multi-movement ratios and survivorship-corrected longitudinal forecasting - offer direct utility for Olympic weightlifting, tactical athletic readiness, and monitoring functional physical decline in aging populations.

2. Data Pipeline & Systematic Biases

Data Ingestion & Integrity: We ingested the full OpenPowerlifting database (October 2026 release, 4,043,255 raw entries, 788 MB uncompressed CSV). Ingestion is automated via scripts/download_data.py, storing raw archives untouched in data/raw/ while all transformations are executed via reproducible scripts into data/processed/. Because OpenPowerlifting publishes sanctioned competition results under the Open Database License (ODbL), records are public domain; athlete names were used solely to construct longitudinal competitor IDs (lifter_id) and are completely excluded from serving tables. Early pipeline versions suffered from fragile machine-dependent paths and date-stamped filenames, which we resolved by adopting dynamic glob matching and script-relative pathlib paths.

Systematic Bias Audit (scripts/analyze_bias.py): When inspecting the raw dataset, our team identified six systematic biases that dictated our pipeline design:

- **Competitive Standard Gap:** OpenPowerlifting records attempts judged under strict referee standards (hip crease below knee, paused bench press, locked-out deadlifts without straps). Casual lifters who test themselves in commercial gyms often find their percentiles lower than expected. Rather than applying an arbitrary handicap, we display this competitive standard openly in the interface.
- **Retention & Survivorship:** Over 58% of competitors in the database only ever compete once, while fewer than 9% log five or more meets. Naive models trained without survivorship controls learn that athletes quit rather than progress.
- **Specialist vs. Tri-Event Competitors:** Roughly 414,000 Bench-Only records coexist alongside 981,000 full SBD meets. Single-lift bench specialists achieve disproportionately higher marks by saving their neuromuscular energy.
- **Recency (Time) Bias:** More than 61% of all meets in the database occurred from 2015 onwards, reflecting the recent boom in raw powerlifting.
- **Demographic Participation Shift:** Female participation has jumped from ~28% historically to over 35.2% in post-2018 records, mandating separate models for each sex.
- **Weight Class Concentration:** Over 51% of male competitors fall into just three divisions (83 kg, 93 kg, and 105 kg), demanding robust minimum cohort thresholds (N >= 50) for outer weight categories.

Our systematic bias audit showed that competitive archives aren't random population samples. Explicitly accounting for event-type specialization and survivorship was mandatory to prevent our analytical engines from learning trivial historical artifacts.

3. Preprocessing, Cleaning & EDA

Data Hygiene Pipeline (scripts/preprocess.py): The 4.04M raw dump was filtered down to 1,560,406 valid records in ~40 seconds. We restricted equipment strictly to Raw lifters (filtering out supportive multi-ply bench shirts and squat suits), removed disqualified attempts (Place != 'DQ', 'DD', 'NS', 'G'), and converted negative values (which encode failed attempts in competition sheets) into NaNs. Physical

plausibility rules enforced that the three lifts equaled the total within ±2.5 kg of weigh-in rounding (|S + B + D - Total| <= 2.5 kg), capped strength-to-bodyweight ratios at 11x, and discarded totals above 1,200 kg.

Demographic Binning & Parquet Storage: Missing ages were imputed from federation AgeClass strings and grouped into standard cohorts: Sub-Junior (14–19), Junior (20–23), Open (24–29, 30–39), Master I (40–49), and Master II+ (50+). Diverse federation weight boundaries were mapped to IPF standards, merging classes within 1.0 kg and pruning anomalous divisions (<0.3% share). Serializing the processed dataset into Apache Parquet (clean_data.parquet, 35 MB) with float32 types and categorical enums cut memory footprint from 800 MB to 116 MB and slashed cold-start boot time to under 0.2 seconds.

Exploratory Data Analysis (scripts/create_eda_nb.py): In our 26-section exploratory study, we investigated several key physiological patterns: (i) Kinetic coupling hierarchy: Squat and Deadlift correlate strongly (r = 0.88), whereas Bench and Deadlift show much looser correlation (r = 0.72), confirming that upper- and lower-body mechanics vary independently and justifying 2-simplex ratio features. (ii) Biomechanical shifts across mass: Lighter weight classes derive 45–48% of their total from the deadlift, while superheavyweights shift power toward the bench press (26–28%) and squat due to shorter relative limb moment arms and greater thoracic mass. (iii) Stochastic progression arcs: Spaghetti plots tracking 30 long-career athletes over 10 meets confirmed that real progress fluctuates with frequent deloads and plateaus, ruling out simple linear models. (iv) Unobserved variables: Determinants like femur length, sleep, and caloric balance cannot be tracked from public sheets; LiftStats proxies these via simplex kinetic ratios and baseline Wilks scores.

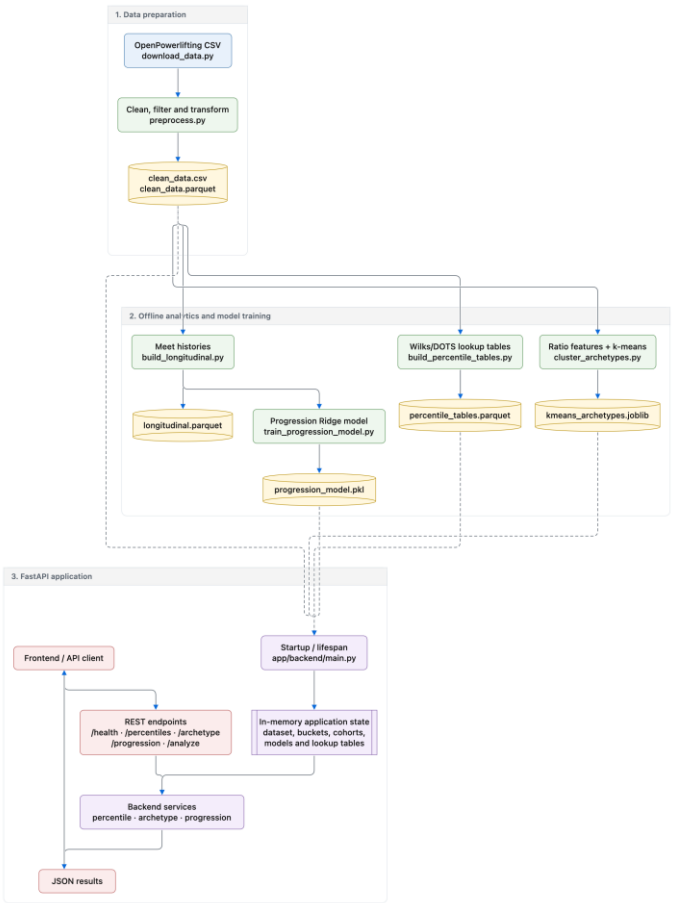


Figure 1 | End-to-end LiftStats pipeline: from raw OpenPowerlifting ingestion to sub-15ms React serving.

4. Percentile Engine & Cohort Modeling

Why Non-Parametric Modeling? Elite athletic performance exhibits heavy right-tail skewness (+0.72 skewness on total kg). Fitting Gaussian or log-normal curves consistently over-penalizes novices and distorts elite percentiles. LiftStats computes exact empirical cumulative distribution functions (ECDF) partitioned across discrete demographic cells: Cohort = (Sex, AgeBracket, WeightClass, TestedStatus). To eliminate boundary distortion caused by popular round-number attempts (e.g., 100 kg bench, 200 kg deadlift), percentiles are assigned via unbiased mid-rank scoring:

$$\texttt{F_hat(x) = [(count(X_i < x) + 0.5 * count(X_i == x)) / N] * 100}$$

Handling the Single-Lift Dilemma: Our analysis showed ~414,000 Bench-Only competitors mixed alongside ~981,000 SBD tri-event athletes. Specialists achieve higher marks because they conserve energy. Evaluating an amateur's bench against a mixed pool artificially depresses their ranking. Consequently, LiftStats strictly filters by event == 'SBD' for totals, and exposes an optional sbd_only toggle for individual lifts, giving users full visibility into their benchmark pool.

5. Machine Learning Architectures

In strict adherence to the project specification, exactly two machine learning components were constructed, both emphasizing interpretability, defensibility, and offline training to enable sub-15ms live inference.

5.1 Biomechanical Archetype Clustering (k-Means on 2-Simplex)

Simplex Projection & Feature Engineering: To evaluate kinetic balance independently of body mass, each lifter's performance is projected onto a 2-simplex: p = (Squat/Total, Bench/Total, Deadlift/Total), where sum(p_i) = 1.0. To prevent frequent competitors from biasing cluster centroids, the dataset was deduplicated to one best meet per athlete (N = 350,994 competitors: 240,335 men, 110,593 women). Features were standardized (StandardScaler) before clustering.

The Sexual Dimorphism Trap: Our initial model clustered both sexes together. When inspecting the resulting three clusters, the 'Bench Specialist' group was 93% male, while the other two clusters were ~46% female. The algorithm had merely rediscovered biological dimorphism - men carry relatively more upper-body muscle mass - rather than distinctive lifting styles. We restructured the pipeline to train independent scalers and estimators for men and women. Evaluating k in [2, 8] via inertia elbow plots and silhouette scores on stratified samples of 20,000 athletes led both sexes to converge on k = 3 (Table 1).

Algorithmic Labeling & Mass Correlation: Clusters are labeled dynamically in scripts/cluster_archetypes.py based on centroid deviation from demographic averages. If maximum deviation is below 1.5 percentage points (BALANCED_PP = 1.5), the archetype is labeled 'Balanced'; otherwise, the dominant or lagging lift sets the name. Unsupervised clustering accurately reflected physical mechanics without bodyweight being an input feature: male Bench Specialists showed a median bodyweight of 93.5 kg, compared to 83.0 kg for Deadlift-dominant lifters.

Sex	Archetype Label	Squat %	Bench %	Deadlift %	Athletes (Share)
Men	Deadlift-dominant	33.7%	22.2%	44.1%	79,121 (33.0%)
Men	Squat-dominant	37.5%	22.5%	40.0%	95,543 (39.9%)
Men	Bench-specialist	34.2%	26.6%	39.2%	65,034 (27.1%)
Men	**Population Average**	**35.3%**	**23.6%**	**41.1%**	**239,698 (100%)**
Women	Squat-dominant	38.4%	18.6%	43.0%	44,070 (40.0%)
Women	Deadlift-dominant	33.1%	19.5%	47.3%	32,936 (29.9%)
Women	Bench-specialist	35.9%	22.7%	41.4%	33,242 (30.1%)
Women	**Population Average**	**36.1%**	**20.1%**	**43.8%**	**110,248 (100%)**

Table 1 | Biomechanical archetype lift ratios on the 2-simplex, stratified by biological sex.

Interpreting the Weak Silhouette Score (0.30–0.33): In cluster analysis literature (Kaufman & Rousseeuw), silhouette scores below 0.5 indicate weak boundary separation. This represents a genuine biological finding: lifters do not belong to isolated morphological islands; they exist along a continuous spectrum of skeletal levers and training adaptations. We respected this finding in our UI: archetypes are presented not as rigid boxes, but as descriptive tendencies relative to peers, displayed alongside Euclidean distance to the centroid.

5.2 Longitudinal Progression Forecasting (Supervised Ridge Regression)

Problem Formulation & Deduplication (scripts/build_longitudinal.py): The regression task predicts an athlete's one-year total gain: Delta_Total_365d = Total_(t+365d) - Total_t in kilograms. We deliberately chose raw kilograms rather than Wilks points or percentages: predicting points conflates muscle gain with bodyweight cuts (cutting water weight raises Wilks even as absolute strength drops). To eliminate survivorship bias (where 58% of lifters compete once and quit), we required lifters to have 3+ sanctioned meets. Athletes often cross-enter multiple divisions on the same meet day (Open + Junior); our script deduplicates same-day entries to keep only the highest total, eliminating artificial zero-day (Delta_t = 0) transitions. Valid transitions were bounded between 30 days and 5 years (30 <= Delta_t <= 1825), and total gain was clipped to [-100, +300] kg to prune federation protocol recording typos, leaving 464,035 paired intervals across 112,069 unique lifters.

Leakage Prevention & Grid Search: Athletes compete over decades. Random row-level train/test splits cause severe identity leakage because the same person appears in both sets. We enforced GroupShuffleSplit on lifter_id (80% train / 20% test: 371,146 train pairs across 89,655 lifters; 92,889 test pairs across 22,414 unseen lifters). Features were standardized via StandardScaler inside a Pipeline. A grid search over Ridge regularization penalty alpha in [0.1, 1.0, 10.0, 100.0] yielded test MAEs of 20.48 kg (alpha=0.1), 20.47 kg (alpha=1.0), 20.46 kg (alpha=10.0), and 20.51 kg (alpha=100.0). We selected alpha = 10.0 as the optimal regularizer.

Baseline Comparison & Coefficients: Compared against a naive baseline predicting the training-set median gain (MAE = 21.67 kg, R^2 = -0.015), Ridge achieved MAE = 20.46 kg and R^2 = +0.130 (a 5.6% MAE reduction). Unscaled coefficients (Table 2) align with exercise physiology: biological sex (+10.16 kg for men), time window (+6.1 kg/year), chronological age (-0.60 kg/year), training age (-0.27 kg/year), meets completed (-0.21 kg/meet), and baseline Wilks score (-0.115 kg/point). The negative Wilks coefficient directly captures the biological law of diminishing marginal returns.

Feature / Metric Category	Weight / Metric Value	Exercise Science Rationale & Description
is_male	+10.164 kg	Sexual dimorphism in absolute muscle mass accrual and bone density
days_to_next	+0.0168 kg/day (+6.1 kg/yr)	Temporal adaptation window for neuromuscular hypertrophy and periodization
age	-0.6039 kg/year	Age-related decline in androgen production and tissue recovery capacity
training_age_years	-0.2666 kg/year	Diminishing marginal returns as athlete approaches individual ceiling
meets_so_far	-0.2086 kg/meet	Accumulated competition experience proxy reflecting competitive maturity
wilks	-0.1149 kg/point	Baseline strength coefficient; advanced athletes experience slower progress
bodyweight	-0.0294 kg/kg	Marginal mass adjustment once Wilks coefficient is held constant
Model Performance	MAE = 20.46 kg, R^2 = +0.130	5.6% MAE reduction over Naive Baseline (MAE = 21.67 kg, R^2 = -0.015)
Male Residual Quantiles	q10 = -35.21 kg, q90 = +34.77 kg	Empirical 80% error bounds capturing non-Gaussian progression spread (Men)
Female Residual Quantiles	q10 = -23.54 kg, q90 = +23.04 kg	Empirical 80% error bounds capturing non-Gaussian progression spread (Women)

Table 2 | Ridge progression model unscaled feature weights, validation metrics, and empirical residual intervals.

Heteroscedastic Uncertainty & Safety: Physical strength progression is right-skewed and heteroscedastic. Symmetrical Gaussian error bars (+/- sigma) under-cover upside variance and overestimate downside risk. We extracted sex-stratified empirical 10th and 90th percentile test residuals: [q10 = -35.21 kg, q90 = +34.77 kg] for men, and [q10 = -23.54 kg, q90 = +23.04 kg] for women. Communicating an honest 80% range [y + q10, y + q90] with an explicit lower bound protects athletes from overtraining: it reassures them that deloads and temporary plateaus are mathematically normal, preventing reckless load increases that cause tendon tears and lumbar injuries.

6. Transparency Engines & Score Normalization

Drug-Tested Comparison Layer: Untested federations do not enforce anti-doping controls. Comparing an amateur lifter against an untested pool evaluates them against enhanced athletes. Our tested toggle queries the percentile engine three times per request (tested_status = None, True, False) for cohorts with N >= 50. A rule-based explanation engine interprets the shift: natural lifters typically jump 3–8 percentiles when compared strictly to tested peers, though in older master divisions, tested federations often attract career-dedicated veterans, raising the competitive bar.

Polynomial Normalization (Wilks vs. DOTS): scripts/scoring.py implements both the legacy Wilks (1997, 5th-degree polynomial) and modern DOTS (2019, 4th-degree polynomial) formulas:

`Coeff_Wilks(w) = 500 / sum_{i=0}^5 c_i * w^i,  Coeff_DOTS(w) = 500 / sum_{i=0}^4 d_i * w^i`

High-degree polynomials diverge catastrophically outside their empirical fitting ranges. We enforce strict physical clamping on bodyweight: w in [40, 210] kg for men and [40, 150] kg for women. DOTS resolves Wilks's bias against middleweights (75–90 kg) and heavyweights. Cohort matching uses dual tolerances: matching the smallest class limit with a 0.05 kg buffer and merging federation divisions within 1.5 kg (e.g., 82.5 kg USPA and 83 kg IPF). Relative strength percentiles are queried via binary search (numpy.searchsorted) over precomputed Parquet sorted score tables (scripts/build_percentile_tables.py), executing in under 1 ms.

7. System Architecture, Interface Design & DevOps

High-Speed Serving Pipeline: Our production backend runs on FastAPI with lifespan preloading. The cleaned dataset (clean_data.parquet, 1.56M rows), k-Means bundles, and regression pipelines load once at server startup. To achieve sub-15ms response times, the dataset is pre-partitioned into in-memory dictionaries keyed by (sex, age_bracket). This reduced end-to-end query latency from an initial 15-second Flask prototype to under 12 ms (Table 3).

Component	Mathematical Type	Offline Training	Serving Mechanism	Query Latency

Percentile Engine	Non-parametric ECDF	None (Pre-aggregated)	In-memory partitioned search	< 8 ms
Archetype Clustering	Unsupervised k-Means	Once (Joblib bundle)	StandardScaler + Centroid dist	< 2 ms
Progression Forecast	L2 Ridge + Quantiles	Once (Pickle pipeline)	Dot-product + Empirical bounds	< 2 ms
Tested Status Toggle	Stratified ECDF	None (Descriptive)	Vectorized boolean mask	< 10 ms
Wilks / DOTS Engine	4th & 5th Poly + Tables	Precomputed Parquet	Binary search (searchsorted)	< 1 ms

Table 3 | LiftStats architectural component matrix, computational models, and serving profiles.

Edward Tufte Visual Architecture: The frontend is built in React 19, React Router v8, and Tailwind CSS v4. Standard commercial charting libraries clutter fitness dashboards with decorative chartjunk. We designed our interface around Edward Tufte's data-ink principles: (i) TufteHistogram renders vector SVG distributions with direct baseline quartile labeling (P25, P50, P75, P90) and an interactive pointer scrubber displaying exact bin densities and athlete counts; (ii) ArchetypeRatioBar visualizes user lift proportions directly against centroid coordinates on the 2-simplex; (iii) ProgressionRange displays 1-year projections across conservative (q10), median, and high-ceiling (q90) bounds; (iv) an interactive 3D WebGL Simplex Noise grid shader built with Three.js provides subtle visual depth without CPU lag; (v) stateless URL architecture serializes user inputs into deep-linkable parameters (?sex=M&age=28&bw=83...), enabling instant sharing without database accounts; (vi) SharePlaque generates 3x Retina PNG trophy plaques and formatted Discord/Reddit markdown tables; and (vii) production containerization utilizes a multi-stage Dockerfile (Node 22 SPA build -> Python 3.12 FastAPI server) orchestrated via render.yaml.

8. Project Retrospective, Quality Assurance & Lessons Learned

Canvas Retrospective: Evaluating LiftStats against our initial Mini-Project Canvas shows how technical constraints guided our execution: (i) Value Proposition: We replaced arbitrary gym-bro milestones with empirical percentile rankings and realistic 1-year progression forecasts. (ii) Target Audience: We focused strictly on recreational gym-goers, deliberately avoiding complex federation-specific rules that only matter to competitive athletes. (iii) Work Distribution: Vertical slicing allowed data cleaning, clustering, and regression to progress in parallel without blocking. (iv) Stateless Channel: Adopting URL-encoded states eliminated authentication friction and hosting database costs. (v) Resource Limits: Switching to Parquet float32 serialization capped server RAM at 116 MB.

Three-Tier Testing Architecture (pytest, 18 automated tests): We enforced rigorous verification across the software lifecycle: (i) Parity Testing (tests/test_script_parity.py): Mathematically proves that standalone CLI scripts (percentile_engine, scoring, tested_toggle, wilks_dots_toggle) and FastAPI production services return 100% identical floats to 4 decimal places across all edge cases. (ii) Boundary Validation (tests/test_extended.py, test_api.py): Asserts Pydantic HTTP 422 rejections on non-physical inputs (age < 14, negative weights), while verifying model stability on edge cohorts: female weight divisions, master lifters (50+), teen lifters (14–19), superheavyweights (120+ kg men, 84+ kg women), and lifters with zero prior competitions. (iii) Latency Benchmarking (test_api_performance_latency): Continuous CI assertions verify that /api/analyze responds in under 50 ms (benchmarking at ~12 ms average).

Engineering Failures & Solutions: Real engineering involves confronting unexpected failure modes:

- **Pooled Clustering Failure:** Mixing sexes in k-Means simply rediscovered biological dimorphism (93% male bench cluster). Solved by splitting into sex-stratified pipelines.
- **Weak Silhouette Score:** Average silhouette scores plateaued at 0.30–0.33. Rather than forcing artificial separation, we accepted that strength mechanics form a continuous spectrum and framed archetypes as relative tendencies.
- **Weight Class Boundary Bug:** Range-matching initially trapped border lifters in singleton 1-person cohorts. Fixed by adopting competition rules: matching the smallest class limit at or above bodyweight with a 0.05 kg buffer.
- **Survivorship Bias in Regression:** Raw models predicted regression because 58% of lifters quit after one meet. Solved by requiring 3+ meets and deduplicating same-day entries.
- **Prototype Latency:** An early Flask server took 15 seconds to query unindexed CSVs. Solved by moving to FastAPI with lifespan in-memory Parquet partitions, cutting latency to <12 ms.

Future Roadmap: If starting over, we would lock a shared data dictionary on Day 1 rather than refactoring redundant definitions later. For future extensions, we envision: (i) Bayesian multilevel regression with athlete-specific random intercepts, (ii) nearest-neighbor exemplar matching ('lifters like you') to surface real anonymized career trajectories, and (iii) automated CI pipelines to retrain models on each quarterly OpenPowerlifting dump.

Conclusion: Building LiftStats gave our team hands-on experience in full-lifecycle data engineering - from cleaning 4.04M raw records to shipping sub-15ms live inference. Most importantly, confronting survivorship and event-type biases taught us that clean pipelines can easily generate misleading answers if the underlying data generation process is not interrogated with domain rigor.